

Adoption of Adjunctive Azithromycin for Unscheduled Cesarean Delivery and Postpartum Infections

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IMPORTANCE Cesarean delivery is the most common surgery in the US and the most significant risk factor for postpartum infection. Although trial evidence demonstrated that adjunctive azithromycin use at the time of birth reduces postpartum infection in pregnant persons who undergo an unscheduled cesarean delivery, use of azithromycin outside that trial has not been reported.

OBJECTIVE To determine the adoption of trial evidence in clinical practice and describe postpartum infection trends among US pregnant persons undergoing unscheduled cesarean delivery.

DESIGN, SETTING, AND PARTICIPANTS Data from Epic Cosmos were used to conduct a difference-in-differences analysis including pregnant individuals who received prenatal care, labored, and gave birth to a liveborn singleton infant at 24 to 43 weeks' gestation from 2013 to 2024. Cesarean birth was considered the treatment group and vaginal birth was considered the control group. The analysis was conducted from July 2025 to June 2026.

EXPOSURE Relative to the publication of trial evidence in 2016, births that occurred from January 1, 2013, to September 28, 2016, were compared with births that occurred from January 1, 2017, to December 31, 2024.

MAIN OUTCOMES AND MEASURES The primary outcomes were perioperative administration of azithromycin and postpartum infection within 6 weeks of delivery. Patient-level linear regression was used to obtain the difference-in-differences estimate.

RESULTS Of the 1 663 441 infant deliveries included in the analysis, there were 202 234 cesarean births and 1 461 207 vaginal births (mean age of pregnant individuals, 29.3 [SD, 5.7] years; median body mass index at time of delivery, 31.8 [IQR, 28.2-36.5]). From January 1, 2013, to September 28, 2016, azithromycin was administered during 2.2% of cesarean births (721/32 970) vs 0.01% of vaginal births (15/261 624) compared with 39.6% (67 094/169 264) vs 0.04% (421/1 199 583), respectively, from January 1, 2017, to December 31, 2024 (adjusted difference-in-differences estimate, 37.6 percentage points [95% CI, 33.1 to 42.2]). From January 1, 2013, to September 28, 2016, postpartum infection occurred after 9.2% of cesarean births (3048/32 970) vs 2.0% of vaginal births (5311/261 624) compared with 8.0% (13 555/169 264) vs 2.7% (32 581/1 199 583), respectively, from January 1, 2017, to December 31, 2024 (adjusted difference-in-differences estimate, -2.0 percentage points [95% CI, -2.6 to -1.4]).

CONCLUSIONS AND RELEVANCE After publication of the trial evidence in 2016, azithromycin use for pregnant persons undergoing unscheduled cesarean delivery increased and postpartum infection rates decreased in the US.

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 Editorial

 Supplemental content

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Postpartum infection is a major cause of maternal morbidity and mortality and is associated with racial, ethnic, and socioeconomic disparities.¹⁻³ In the US, nearly one-third of births occur by cesarean delivery, which remains the most important risk factor for postpartum infection.^{2,4} Overall infection rates after cesarean delivery range from 10% to 20%, and are nearly 2-fold higher in persons undergoing unscheduled cesarean delivery compared with those undergoing scheduled cesarean delivery.^{2,5,6} In 2024, there were more than 1 million births with cesarean delivery, and approximately 30% were unscheduled (occurred after a trial of labor), suggesting more than 60 000 pregnant individuals will develop a postpartum infection annually.⁷

A double-blind, pragmatic, randomized clinical trial (the Cesarean Section Optimal Antibiotic Prophylaxis trial⁸) was published in 2016 that compared postpartum infection rates among 2013 women undergoing an unscheduled cesarean delivery (were randomized to adjunctive intravenous azithromycin or placebo). In that trial,⁸ all participants received routine preincision surgical antibiotic prophylaxis and those who received standard group B streptococcus antibiotic prophylaxis were not excluded. The trial⁸ demonstrated a 50% reduction in postpartum infection in the azithromycin group (6% vs 12% in the placebo group). Follow-up studies have supported the intervention's fetal safety and cost-effectiveness.^{9,10} The American College of Obstetricians and Gynecologists endorsed the use of azithromycin in a 2018 practice bulletin update,¹¹ noting that it may be "considered" for unscheduled cesarean deliveries.

The extent to which azithromycin was adopted into clinical practice after publication of the trial⁸ results is unknown. Moreover, there are often concerns¹² about the generalizability of trial findings to heterogeneous populations that may not resemble the study participants. Thus, the objective of this study was to characterize the uptake of azithromycin for unscheduled cesarean delivery in the US after publication of the trial evidence,⁸ and determine whether there was an associated decrease in postpartum infection rates.

Methods

Observational data from Epic Cosmos (Epic Systems Corporation) for births occurring between January 1, 2013, and December 31, 2024, were used in this difference-in-differences analysis. The Epic Cosmos dataset was created in collaboration with health systems in the US and includes more than 300 million patients who received treatment at nearly 2000 hospitals and 45 000 clinics (as of March 2026¹³) located throughout all 50 states and Washington, DC. As use of the Epic system has grown over time, so has the number of sites contributing patient data to Cosmos. Therefore, not all institutions contributed data consistently across the study period.

This study was classified as non-human participants research and was exempt from institutional review board approval and informed consent. This study used the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.¹⁴

Key Points

Question Among US pregnant persons who underwent an unscheduled cesarean delivery, did azithromycin use and postpartum infection rates change after publication of trial evidence in 2016?

Findings In the adjusted difference-in-differences analysis of 1.7 million births, azithromycin use increased by 37.6 percentage points for cesarean births relative to vaginal births and postpartum infections decreased by 2.0 percentage points for cesarean births relative to vaginal births after publication of the trial.

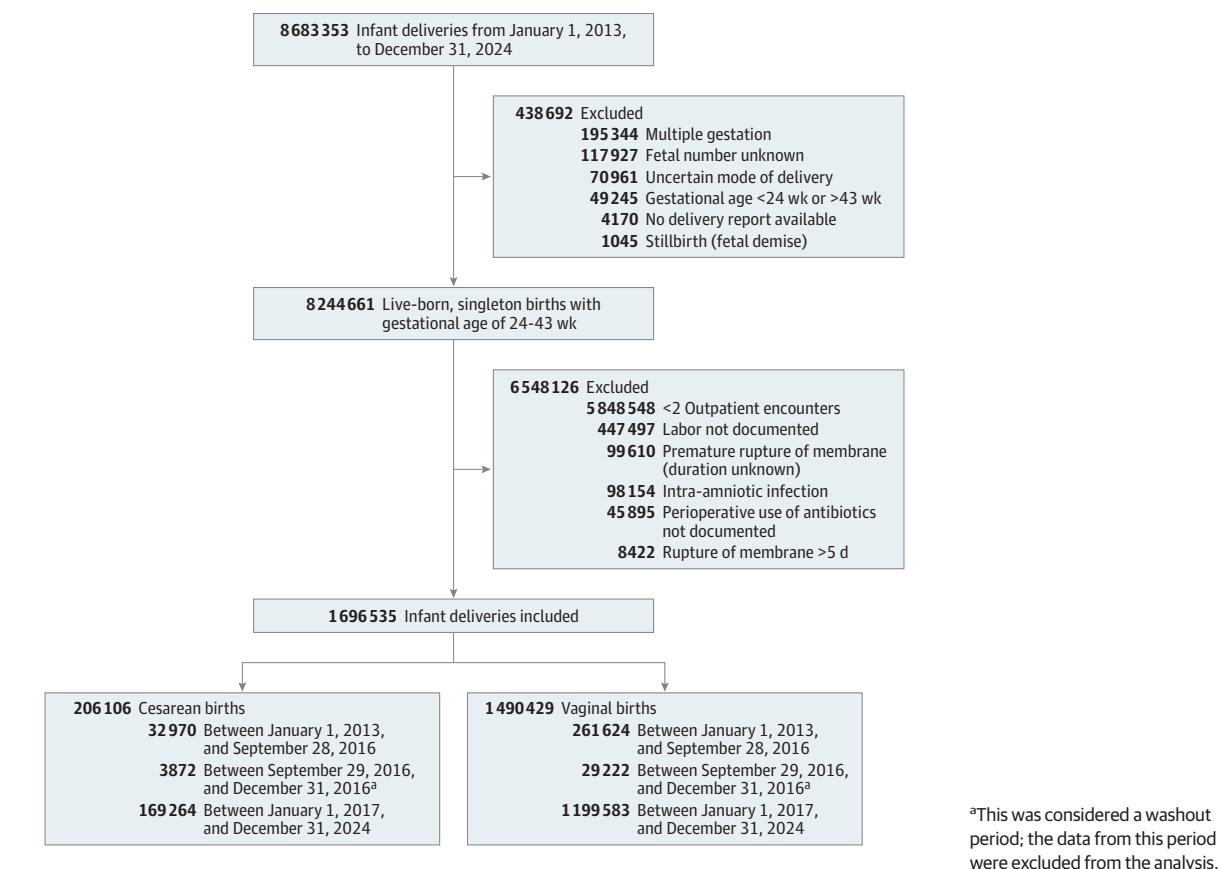
Meaning After publication of the trial, azithromycin use increased and postpartum infections decreased among pregnant persons who underwent an unscheduled cesarean delivery in the US outside the clinical trial setting.

We included pregnant individuals who labored and gave birth to a liveborn singleton infant at 24 weeks 0 days' gestation to 43 weeks 6 days' gestation between 2013 and 2024 (Figure 1). The characteristics of labor were based on information recorded in the Epic Cosmos dataset and included labor type (induction or spontaneous), start of labor, length for first stage of labor, start time or length for second stage of labor, whether forceps or vacuum delivery was attempted, and mode of delivery. Similar variables have been used to identify attempted vaginal birth among cesarean births in prior literature.^{15,16} Individuals with a rupture of membrane to delivery duration of 4 hours or longer also were included regardless of labor status, which is consistent with the eligibility criteria of the 2016 trial.⁸

We excluded individuals if they did not have at least 2 outpatient visits in an obstetric, midwifery, or family medicine department during the pregnancy; this requirement was set to increase the likelihood of ascertaining postpartum infectious outcomes that may be managed in an office setting (eg, wound infection). Consistent with the 2016 trial,⁸ we excluded individuals with an encounter diagnosis for intra-amniotic infection during the delivery encounter (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision [ICD-10]* code O41.1). In addition, we excluded individuals whose mode of delivery (vaginal or cesarean) was unknown and those with a duration for rupture of membrane of longer than 5 days (as a proxy for preterm, prelabor rupture of membrane) or with an unknown duration. We also excluded individuals without perioperative antibiotics documented because this would reflect standard of care was not followed, and those with only azithromycin documented at the time of cesarean delivery because this would not constitute adjunctive use.

For the difference-in-differences analysis, we used 2 periods relative to the September 29, 2016, publication date of the trial.⁸ January 1, 2013, to September 28, 2016, was used as the period prior to publication of the trial.⁸ September 29, 2016, to December 31, 2016, was used as a washout period; individuals delivering during this period were excluded to reduce the influence of early adoption of the trial evidence. January 1, 2017, to December 31, 2024, was used as the period after publication of the trial.⁸ We extended the latter period through the

Figure 1. Flow Diagram Depicting the Study Inclusion and Exclusion Criteria



end of 2024 to enable a contemporary analysis that also accounted for increasing adoption of the trial evidence over time. As a sensitivity analysis, we varied the duration of the latter period.

Because the trial findings were only relevant in individuals undergoing cesarean delivery,⁸ the current analysis considered individuals who delivered by cesarean to be the treatment group and those who delivered by vaginal delivery to be the control group. Vaginal births were selected as the control group for 2 reasons. The type of cesarean birth (scheduled or unscheduled) was not reliably available for the cohort. Although proxy measures were used to determine labor status, we could not be certain that their absence indicated a scheduled cesarean birth. In addition, some institutions have adopted azithromycin for scheduled cesarean birth, potentially limiting its use as an unexposed comparator.¹⁷

Patient Characteristics

The patient characteristics were obtained from the electronic health record. The following characteristics were compared between the periods for the treatment group vs the control group: maternal age, maternal race, Hispanic ethnicity, parity, body mass index (calculated as weight in kilograms divided by height in meters squared; categorized as <18.5, 18.5-24.9, 25-29.9, 30-39.9, or ≥40) at time of delivery, prior cesarean birth, gestational age at delivery (categorized as 24-27, 28-31, 32-33, 34-

36, 37-38, or ≥39 weeks), pregestational diabetes or gestational diabetes requiring medication (as diagnosed by at least 2 encounter ICD-10 codes; eTable 1 in Supplement 1), duration for rupture of membrane to delivery was 18 hours or longer, reached second stage of labor, and insurance type.

Race is reported in the Cosmos dataset as 6 individual race categories (American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White, or other [not represented by the 5 preceding categories]) or multiracial (if >1 race reported). Patient demographics, including race and ethnicity, were examined to compare the patient characteristics in this study with those in the 2016 trial.⁸ Delivery location also was obtained from a delivery department identifier.

Outcomes

The primary outcomes of interest were (1) perioperative administration of azithromycin and (2) postpartum infection (composite outcome) within 6 weeks of delivery. The timing of intravenous azithromycin administration was identified in the electronic health record and ranged from 90 minutes before birth to 60 minutes after birth. The time of birth was used because incision time is unavailable in the dataset. In the 2016 trial,⁸ azithromycin was infused over 60 minutes and was administered before incision or as soon as possible thereafter. We included a 60-minute window after birth to account for delayed

administration or discrepancy between administration and documentation time. Postpartum infection (composite outcome) included any of the following ICD-10 diagnoses: endometritis, meningitis, other postpartum infection, pneumonia, pyelonephritis, sepsis, urinary tract infection, wound disruption, or wound infection (eTable 1 in Supplement 1).

As a sensitivity analysis, we also examined the postpartum infection outcome with exclusion of the individual components of urinary tract infection and wound disruption. These individual components were not included in the 2016 trial,⁸ but were included in our analysis to account for possible overlap with more significant infections, given the use of ICD-10 codes. Individual categories were mutually exclusive, although an individual delivery could have multiple associated infections.

Statistical Analysis

The difference-in-differences analysis used a patient-level linear regression model. Based on a priori hypotheses and published risk factors for infection during labor,¹⁸⁻²¹ the adjusted models controlled for age, parity, body mass index at delivery, diabetes, prior cesarean birth, gestational age, and whether duration from rupture of membrane to delivery was 18 hours or longer. Differences in the patient-level characteristics before and after the 2016 trial⁸ also were assessed. Individuals with missing covariate data were excluded from the adjusted analysis. Estimated marginal means were calculated by generating predicted values for each treatment observation in the dataset and by averaging across all observations. The parallel trends assumption did not appear to be violated based on a visual inspection of period trends (before the 2016 trial⁸), event study plots, and by fitting a linear model to the before period and testing for statistical significance of the time × treatment group interaction (eTable 2 and eFigure in Supplement 1).

Several additional sensitivity analyses were performed. To account for site-specific changes over time, an additional model included site × period fixed effects. To decrease reliance on larger sites, we restricted the cohort to sites that contributed fewer than 1000 deliveries during the study period. Another analysis included deliveries to individuals without outpatient care in the system.

Data extraction was conducted using version 20.2 of SQL Server Studio (Microsoft) on July 26, 2025. The statistical analyses were conducted from July 2025 to June 2026 and performed using RStudio version 4.5.1 (Posit Software, PBC). Standardized mean differences were calculated for univariate analyses; values less than 0.10 were considered evidence of balance. A 2-sided $P < .05$ was considered statistically significant. Robust SEs were clustered by delivery department and by individual to account for within-site correlation and for individuals with more than 1 delivery during the study period.

Results

Of the 8 683 353 deliveries with a recorded birth during the study period, 1 696 535 (19.5%) met the study criteria (Figure 1). There were 294 594 deliveries from January 1, 2013, to

September 28, 2016, 33 094 from September 29, 2016, to December 31, 2016 (washout period), and 1 368 847 from January 1, 2017, to December 31, 2024. The births that occurred during the washout period were excluded from the analysis. A total of 1 663 441 deliveries were included in the final analysis and were attributed to 1 350 238 unique individuals. In the final cohort, there were 202 234 cesarean deliveries (12.2%). Births were reported in 1951 unique delivery departments. The characteristics of the study population appear in Table 1.

Primary Outcomes

After 2016, the rate of perioperative azithromycin use increased significantly among cesarean births (Figure 2). From January 1, 2013, to September 28, 2016, azithromycin was administered during 2.19% (95% CI, 2.03% to 2.35%) of cesarean births (721 of 32 970) vs 0.01% (95% CI, 0% to 0.01%) of vaginal births (15 of 261 624) compared with 39.6% (95% CI, 39.4% to 39.9%; 67 094/169 264) vs 0.04% (95% CI, 0.03% to 0.04%; 421/1 199 583), respectively, from January 1, 2017, to December 31, 2024 (adjusted difference-in-differences estimate, 37.6 percentage points [95% CI, 33.1 to 42.2]) (Table 2).

From January 1, 2013, to September 28, 2016, postpartum infection occurred after 9.2% (95% CI, 8.9% to 9.6%) of cesarean births (3048/32 970) vs 2.0% (95% CI, 2.0% to 2.1%) of vaginal births (5311/261 624) compared with 8.0% (95% CI, 7.9% to 8.1%; 13 555/169 264) vs 2.7% (95% CI, 2.7% to 2.8%; 32 581/1 199 583), respectively, from January 1, 2017, to December 31, 2024 (adjusted difference-in-differences estimate, −2.0 percentage points [95% CI, −2.6 to −1.4]; unadjusted difference-in-differences estimate, −1.9 percentage points [95% CI, −2.5 to −1.3]) (Table 2). Compared with the counterfactual rate of 10.0% (obtained from the estimated marginal mean of 8.0% for the adjusted rate for the period from January 1, 2017, to December 31, 2024, and the difference-in-differences estimate of −2.00 percentage points), the adjusted difference-in-differences estimate represents a 20% relative decrease.

The rates of the individual infection components appear in Table 2. Among cesarean deliveries, there were significant reductions across a variety of infections for the period from January 1, 2017, to December 31, 2024, compared with the period from January 1, 2013, to September 28, 2016, including for endometritis, sepsis, and surgical site infections (eg, wound infection or disruption). The rates of urinary tract infection were higher for the period from January 1, 2017, to December 31, 2024, among cesarean births; this trend was also seen among vaginal births. The findings were robust to sensitivity analyses (eTables 3-4 in Supplement 1).

Discussion

After the trial⁸ was published in 2016, perioperative use of azithromycin during an unscheduled cesarean birth increased and associated postpartum infection rates decreased relative to the change in vaginal births. The findings were robust to covariate adjustment and several sensitivity analyses.

Table 1. Baseline Characteristics of the Study Population

	Cesarean birth ^a			Vaginal birth ^a		
	Jan 1, 2013, to Sep 28, 2016 (n = 32 970)	Jan 1, 2017, to Dec 31, 2024 (n = 169 264)	SMD ^b	Jan 1, 2013, to Sep 28, 2016 (n = 261 624)	Jan 1, 2017, to Dec 31, 2024 (n = 1 199 583)	SMD ^b
Demographics						
Age, mean (SD), y	29.6 (6.0)	30.2 (5.9)	0.11	28.7 (5.7)	29.3 (5.7)	0.11
Race ^c	(n = 32 528)	(n = 166 638)		(n = 258 024)	(n = 1 179 856)	
American Indian or Alaska Native	155 (0.5)	705 (0.4)		1359 (0.5)	4816 (0.4)	
Asian	1472 (4.5)	7362 (4.4)		11 725 (4.5)	51 464 (4.4)	
Black or African American	6085 (18.7)	30 496 (18.3)		39 333 (15.2)	183 389 (15.5)	
Multiracial ^d	4952 (15.2)	23 932 (14.4)	0.05	32 391 (12.6)	152 890 (13.0)	0.04
Native Hawaiian or Other Pacific Islander	110 (0.3)	756 (0.5)		1031 (0.4)	5756 (0.5)	
White	17 507 (53.8)	92 693 (55.6)		152 964 (59.3)	701 415 (59.4)	
Other ^e	2247 (6.9)	10 694 (6.4)		19 221 (7.4)	80 126 (6.8)	
Hispanic ethnicity, No./total (%) ^c	5821/31 520 (18.5)	29 022/161 190 (18.0)	0.02	49 366/250 244 (19.7)	215 688/1 144 439 (18.8)	0.03
Type of insurance	(n = 26 168)	(n = 152 467)		(n = 201 738)	(n = 1 070 328)	
Private	13 608 (52.0)	86 975 (57.0)		100 017 (49.6)	571 951 (53.4)	
Public	12 341 (47.2)	63 564 (41.7)	0.32	99 771 (49.5)	483 600 (45.2)	0.34
Other ^f	219 (0.8)	1928 (1.3)		1950 (1.0)	14 777 (1.4)	
Medical history						
Pregestational diabetes	1436 (4.4)	8427 (5.0)	0.03	3645 (1.4)	20 357 (1.7)	0.03
Gestational diabetes (class A2)	339 (1.0)	8801 (5.2)	0.24	1481 (0.6)	39 573 (3.3)	0.20
Body mass index category at delivery ^g	(n = 31 314)	(n = 161 761)		(n = 246 738)	(n = 1 138 729)	
<18.5	<11 (<1.0) ^h	20 (<1.0)		65 (<1.0)	313 (<1.0)	
18.5-24.9	1442 (4.6)	5700 (3.5)		22 606 (9.2)	84 735 (7.4)	
25-29.9	7250 (23.2)	32 234 (19.9)	0.13	84 850 (34.4)	350 943 (30.8)	0.13
30-39.9	15 985 (51.0)	82 598 (51.1)		114 591 (46.4)	557 325 (48.9)	
≥40	6629 (21.2)	41 209 (25.5)		24 626 (10.0)	145 413 (12.8)	
Pregnancy and delivery characteristics						
Nulliparity	18 544 (56.2)	98 805 (58.4)	0.04	94 828 (36.2)	444 554 (37.1)	0.02
Prior cesarean birth	8836 (26.8)	43 045 (25.4)	0.03	10 899 (4.2)	53 160 (4.4)	0.01
Rupture of membrane to delivery ≥18 h	5416 (16.4)	33 157 (19.6)	0.08	16 625 (6.4)	85 830 (7.2)	0.03
Gestational age range, wk ⁱ						
39-43	20 984 (63.6)	99 754 (58.9)		176 784 (67.6)	769 353 (64.1)	
37-38	8827 (26.8)	51 805 (30.6)		69 231 (26.5)	358 093 (29.9)	
34-36	2469 (7.5)	14 382 (8.5)		13 537 (5.2)	63 448 (5.3)	
32-33	354 (1.1)	1784 (1.1)	0.10	1174 (0.4)	5256 (0.4)	0.08
28-31	227 (0.7)	1022 (0.6)		653 (0.2)	2531 (0.2)	
24-27	109 (0.3)	517 (0.3)		245 (0.1)	902 (0.1)	

Abbreviation: SMD, standardized mean difference.

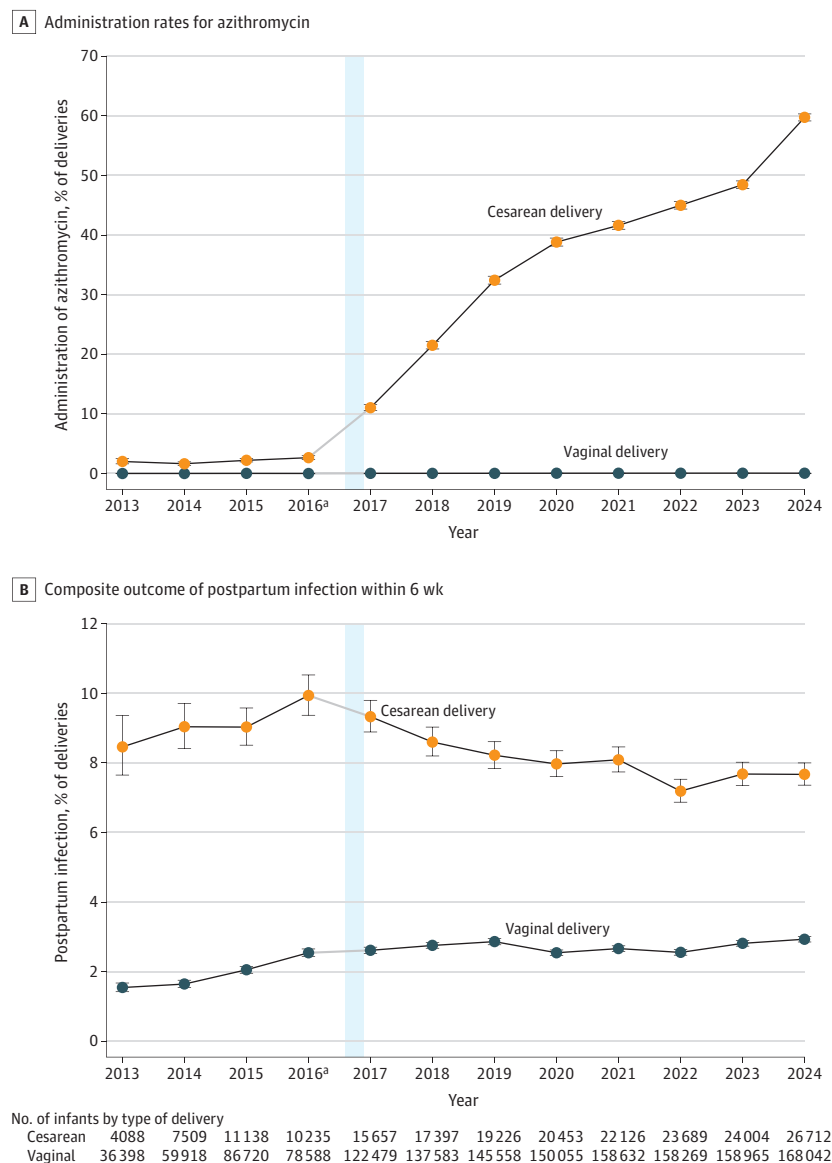
^a Data are expressed as No. (%) unless otherwise indicated. Some individuals contributed multiple deliveries to the study population.^b Calculated by averaging pairwise comparisons across the group. An SMD less than 0.10 was considered evidence of balance.^c As reported in Epic Cosmos from closed categories.^d More than 1 race reported.^e Selected by patients who felt they did not fall into the other provided categories.^f Workers' compensation, third-party liability, or reported as "other" in Epic Cosmos.^g Calculated as weight in kilograms divided by height in meters squared.^h Suppressed per Epic Cosmos guidelines.ⁱ Excluded deliveries that occurred before the gestational age of 24 weeks or during the gestational age of 44 weeks or later.

Although randomized clinical trials are the gold standard for determining the efficacy of an intervention, their translation into clinical practice can be limited by issues of generalizability outside highly controlled environments.^{22,23} The current study suggests that perioperative azithromycin use at the time of unscheduled cesarean delivery is associated with decreased infection in a large, real-world cohort outside the clinical

trial setting that is representative of the US population. These findings support the external validity of the 2016 trial.⁸

Although our study supports the conclusions of the 2016 trial,⁸ there are some notable differences. The postpartum infection rate in the trial⁸ was 6% in those receiving azithromycin vs 12% in the control group. In contrast, we observed a lower postpartum infection rate in the cesarean birth group during

Figure 2. Line Graphs Depicting the Trends for the Primary Outcomes of Azithromycin Administration and Postpartum Infection



^aIncludes period from January 1, 2016, to September 28, 2016.

The blue shaded area indicates the washout period (September 29, 2016, to December 31, 2016). The data from this period were excluded from the analysis (additional information appears in the Methods section). The error bars indicate the 95% CIs.

B, The composite outcome included any of the following (identified from *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision*, codes): endometritis, meningitis, other postpartum infection, pneumonia, pyelonephritis, sepsis, urinary tract infection, wound disruption, or wound infection.

the period from January 1, 2013, to September 28, 2016. This is likely due in part to the use of *ICD-10* codes to identify postpartum infection (including outcome underascertainment and variable sensitivity of individual codes), the use of electronic health record data (which does not capture care outside of the Epic network), and differences in the study populations. For example, our study population had a higher rate of individuals with private insurance (approximately 53% vs 32% in the 2016 trial⁸); having private insurance is associated with lower postpartum infection rates.^{24,25}

We also observed a smaller relative decrease in infection rates after increased azithromycin use (20% vs 49% in the 2016 trial⁸). The lower reduction may be due in part to the fact that less than 50% of individuals received azithromycin during the period from January 1, 2017, to December 31, 2024, com-

pared with 100% in the azithromycin group in the 2016 trial.⁸ Therefore, we would not expect the mean change during the period from January 1, 2017, to December 31, 2024, to mirror that of the intervention group in the trial,⁸ given that practice adoption is increasing over time. In addition, antibiotic use in clinical practice does not adhere to a strict protocol (vs a clinical trial setting). Given that more than 300 000 individuals have an unscheduled cesarean delivery annually, a 20% reduction, if due to the use of azithromycin, would still represent a meaningful improvement in postpartum outcomes, potentially preventing 6000 infections each year.⁷

There are several strengths to our study. This study leverages a large, contemporary, national electronic health record database that includes births from all 50 states across academic and community hospitals in addition to detailed birth

Table 2. Primary Outcomes

	Cesarean birth		Between-period difference (95% CI), %	Vaginal birth		Between-period difference (95% CI)	Difference-in-differences estimate, percentage points (95% CI) ^a	
	Jan 1, 2013, to Sep 28, 2016 (n = 32 970)	Jan 1, 2017, to Dec 31, 2024 (n = 169 264)		Jan 1, 2013, to Sep 28, 2016 (n = 261 624)	Jan 1, 2017, to Dec 31, 2024 (n = 1 199 583)		Unadjusted model	Adjusted model ^b
Primary outcomes, No. (%)								
Perioperative administration of azithromycin	721 (2.2)	67 094 (39.6)	37.5 (37.2 to 37.7)	15 (0.01)	421 (0.04)	0.03 (0.02 to 0.03)	37.4 (32.9 to 41.9)	37.6 (33.1 to 42.1)
Postpartum infection (composite outcome) within 6 wk of delivery	3048 (9.2)	13 555 (8.0)	-1.2 (-1.6 to -0.9)	5311 (2.0)	32 581 (2.7)	0.69 (0.62 to 0.75)	-1.9 (-2.5 to -1.3)	-2.0 (-2.6 to -1.4)
Individual components of postpartum infection, No. (%) ^c								
Wound infection	1214 (3.7)	5277 (3.1)	-0.56 (-0.79 to -0.34)	329 (0.1)	1576 (0.1)	0.01 (-0.01 to 0.02)		
Wound disruption	901 (2.7)	4101 (2.4)	-0.31 (-0.5 to -0.12)	241 (0.1)	1105 (0.1)	0 (-0.01 to 0.01)		
Endometritis	655 (2.0)	2541 (1.5)	-0.49 (-0.65 to -0.32)	1494 (0.6)	7363 (0.6)	0.04 (0.01 to 0.08)		
Sepsis	608 (1.8)	1467 (0.9)	-0.98 (-1.13 to -0.82)	1554 (0.6)	5438 (0.5)	-0.14 (-0.17 to -0.11)		
Other postpartum infection	490 (1.5)	1954 (1.2)	-0.33 (-0.47 to -0.19)	1667 (0.6)	9648 (0.8)	0.17 (0.13 to 0.20)		
Urinary tract infection	222 (0.7)	2089 (1.2)	0.56 (0.46 to 0.67)	1330 (0.5)	13 138 (1.1)	0.59 (0.55 to 0.62)		
Pneumonia	108 (0.3)	445 (0.3)	-0.06 (-0.13 to 0)	288 (0.1)	1316 (0.1)	0 (-0.01 to 0.01)		
Pyelonephritis	37 (0.1)	251 (0.2)	0.04 (-0.01 to 0.08)	319 (0.1)	2083 (0.2)	0.05 (0.04 to 0.07)		
Meningitis	<11 (<1.0) ^d	13 (<1.0)	NA	16 (<1.0)	79 (<1.0)	0 (0 to 0)		

Abbreviation: NA, not available.

^a These estimates were obtained from the linear regression models. The difference-in-differences models were not run for the individual components of postpartum infection (composite outcome).^b The model included age, body mass index at delivery, prior cesarean delivery, nulliparity, gestational age at delivery, prolonged duration for rupture of membrane to delivery, pregestational diabetes, or gestational diabetes (class A2).^c The individual components do not sum to the total number of deliveries for postpartum infection because multiple infections could occur after a delivery.^d Suppressed per Epic Cosmo guidelines.

and prescription data, enhancing the generalizability of our findings. The use of the difference-in-differences method lends robustness to the findings compared with other types of analyses. In particular, the inclusion of a control group (vaginal births) unaffected by the dissemination of the trial⁸ evidence tempers the limitations of differential outcome assessment over time (eg, the *ICD-10* transition that occurred in October 2015). We hypothesize that the increasing rates of postpartum infection seen from 2013 to 2016 were partly due to the increased (and eventually required) use of *ICD-10* codes, which appears supported by the stable rates in the vaginal delivery group after 2016.

Limitations

There are also several limitations to our study. First, vaginal births were used as a control group because we could not identify scheduled cesarean births with certainty, and there is some evidence of azithromycin use for scheduled cesarean births.¹⁷ We cannot account for other interventions that may have been changing during the period from January 1, 2017, to December 31, 2024, to reduce postoperative infection independently of azithromycin use (such as enhanced recovery after surgery pathways or the use of specific abdominal or vaginal preparations).

Second, the Epic Cosmos database does not provide access to all clinical information that is captured during a delivery encounter, so some residual confounding may remain; however, difference-in-differences approaches do account for time-

invariant confounding. Third, patients in Epic Cosmos tend to be slightly older and are more likely to be privately insured relative to the US census, hospitals are more likely to be urban and academic, and we excluded patients without prenatal care in the system; these variables may affect generalizability.²⁶ Fourth, because not all departments reported data equally across the study, differences at individual sites (such as in perioperative protocols or baseline rates) could influence the changes seen over time, although these results were robust to several sensitivity analyses.

Fifth, we mapped infectious complications from the 2016 trial⁸ to corresponding *ICD-10* codes. Specific codes for postpartum infection vary in their accuracy, although we do not hypothesize that accuracy is related to mode of delivery.²⁷ We included a broader range of *ICD-10* codes in a composite outcome because this has been shown to improve sensitivity and reduce the reliance on a single code or encounter.²⁸ Sixth, neither the Cesarean Section Optimal Antibiotic Prophylaxis trial⁸ nor this analysis were designed to measure differences in maternal death.

Conclusions

After publication of the trial evidence in 2016, azithromycin use for pregnant persons undergoing unscheduled cesarean delivery increased and postpartum infection rates decreased in the US.

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